

MongoDB Assignment 1: CRUD Operations and Query Execution

```
// Create/Switch Database
```

```
use infotechdb
```

```
// Insert Documents
```

```
db.infotechdb.insertOne({  
  name: "chaitanya",  
  age: 20,  
  email: "chaitu@infotech.com",  
  course: [  
    { name: "mongodb", duration: 7 },  
    { name: "java", duration: 30 }  
  ]  
})
```

```
db.infotechdb.insertOne({  
  name: "shalini",  
  age: 17,  
  email: "shalini@infotech.com",  
  course: [  
    { name: "basic computer", duration: 30 },  
    { name: "python", duration: 30 }  
  ]  
})
```

```
db.infotechdb.insertOne({  
  name: "haritha",  
  age: 23,  
  email: "haritha@infotech.com",  
  course: [  
    { name: "basic computer", duration: 30 },  
    { name: "python", duration: 30 }  
  ]  
})
```

```
db.infotechdb.insertOne({  
  name: "sagar",  
  age: 20,  
  email: "sagar@infotech.com",  
  course: [  
    { name: "basic computer", duration: 30 },  
    { name: "python", duration: 30 }  
  ]  
})
```

```
// Display All Documents
```

```
db.infotechdb.find().pretty()
```

```
// Insert Multiple Documents
```

```
var data = [  
  {  
    studentID: 1020,  
    student_name: "shalini",  
    age: 17,  
    course: "BCA"  
  },  
  {  
    studentID: 1030,  
    student_name: "subhath",  
    age: 18,  
    course: "BCOM"  
  },  
  {  
    studentID: 1040,  
    student_name: "sagar",  
    age: 20,  
    course: "BBA"  
  }  
]
```

```
db.infotechdb.insertMany(data)
```

```
// Display All Records
```

```
db.infotechdb.find().pretty()
```

```
// Query 1
```

```
db.infotechdb.find({
  student_name: "sagar"
}).pretty()
```

```
// Query 2
```

```
db.infotechdb.find({
  age: { $gt: 17 }
}).pretty()
```

```
// Query 3
```

```
db.infotechdb.find({
  studentID: { $lt: 1030 }
}).pretty()
```

```
// Query 4
```

```
db.infotechdb.find({
  age: { $gte: 17 }
}).pretty()
```

OUTPUT:

```
{
  {
    "_id" : ObjectId("6a436081a27fb3ec480d0523"),
    "name" : "sagar",
    "age" : 20,
    "email" : "sagar@infotech.com",
    "course" : [
      {
        "name" : "basic computer",
        "duration" : 30
      },
      {
        "name" : "python",
        "duration" : 30
      }
    ]
  }
}

{
  "_id" : ObjectId("6a436081a27fb3ec480d0524"),
  "studentID" : 1020,
  "student_name" : "shalini",
  "age" : 17,
  "course" : "BCA"
}

{
  "_id" : ObjectId("6a436081a27fb3ec480d0525"),
  "studentID" : 1030,
  "student_name" : "subhath",
  "age" : 18,
  "course" : "BCOM"
}

{
  "_id" : ObjectId("6a436081a27fb3ec480d0526"),
  "studentID" : 1040,
  "student_name" : "sagar",
  "age" : 20,
  "course" : "BBA"
}
>
```